▶ CARDIOVASCULAR—PHYSIOLOGY

Cardiac output variables

Stroke volume	**S**troke **V**olume affected by **C**ontractility, **A**fterload, and **P**reload. ↑ SV with: ▪ ↑ Contractility (eg, anxiety, exercise) ▪ ↑ Preload (eg, early pregnancy) ▪ ↓ Afterload	**SV CAP**. Stroke work (SW) is work done by ventricle to eject SV. SW ∝ SV × MAP A failing heart has ↓ SV (systolic and/or diastolic dysfunction).

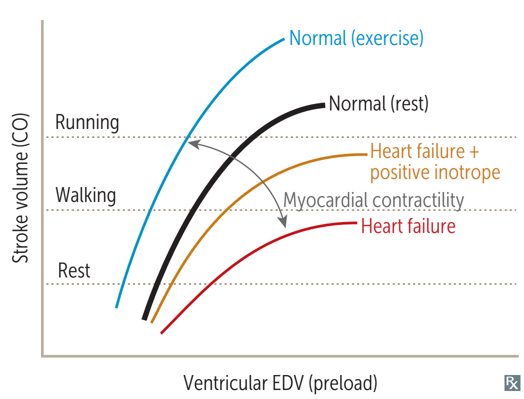
Contractility	Contractility (and SV) ↑ with:
	▪ Catecholamine stimulation via β_1 receptor:
	  ▪ Activated protein kinase A → phospholamban phosphorylation → active Ca^{2+} ATPase → ↑ Ca^{2+} storage in sarcoplasmic reticulum
	  ▪ Activated protein kinase A → Ca^{2+} channel phosphorylation → ↑ Ca^{2+} entry → ↑ Ca^{2+}-induced Ca^{2+} release
	▪ ↑ intracellular Ca^{2+}
	▪ Digoxin (blocks Na^+/K^+ pump → ↑ intracellular Na^+ → ↓ Na^+/Ca^{2+} exchanger activity → ↑ intracellular Ca^{2+})
		Contractility (and SV) ↓ with:
		▪ β_1-blockade (↓ cAMP)
		▪ Heart failure (HF) with systolic dysfunction
		▪ Acidosis
		▪ Hypoxia/hypercapnia (↓ P_{O_2}/↑ P_{CO_2})
		▪ Nondihydropyridine Ca^{2+} channel blockers

Preload	Preload approximated by ventricular end-diastolic volume (EDV); depends on venous tone and circulating blood volume, both of which affect venous return.	Venous vasodilators (eg, nitroglycerin) ↓ preload.
Cardiac oxygen demand	Myo**card**ial O_2 demand is ↑ by: ▪ ↑ **c**ontractility ▪ ↑ **a**fterload (proportional to arterial pressure) ▪ ↑ heart **r**ate ▪ ↑ **d**iameter of ventricle (↑ wall tension) Coronary sinus contains most deoxygenated blood in body.	Wall tension follows Laplace's law: Wall tension = pressure × radius $\text{Wall stress} = \dfrac{\text{pressure} \times \text{radius}}{2 \times \text{wall thickness}}$
Afterload	Afterload approximated by MAP. ↑ pressure → ↑ wall tension per Laplace's law → ↑ afterload. LV compensates for ↑ afterload by thickening (hypertrophy) in order to ↓ wall stress.	**Ar**terial vasodilators (eg, hydr**ala**zine) ↓ **a**fterload. ACE inhibitors and ARBs ↓ both preload and afterload. Chronic hypertension (↑ MAP) → LV hypertrophy.

Cardiac output equations

	EQUATION	NOTES
Stroke volume	SV = EDV – ESV	ESV = end-systolic volume.
Ejection fraction	$EF = \frac{SV}{EDV} = \frac{EDV - ESV}{EDV}$	EF is an index of ventricular contractility (↓ in systolic HF; usually normal in diastolic HF). Normal EF is 50%–70%.
Cardiac output	$CO = \dot{Q} = SV \times HR$ Fick principle: $CO = \frac{\text{rate of } O_2 \text{ consumption}}{(\text{arterial } O_2 \text{ content} - \text{venous } O_2 \text{ content})}$	In early stages of exercise, CO maintained by ↑ HR and ↑ SV. In later stages, CO maintained by ↑ HR only (SV plateaus). Diastole is shortened with ↑↑ HR (eg, ventricular tachycardia) → ↓ diastolic filling time → ↓ SV → ↓ CO.
Pulse pressure	PP = systolic blood pressure (SBP) – diastolic blood pressure (DBP)	PP directly proportional to SV and inversely proportional to arterial compliance. ↑ PP in aortic regurgitation, aortic stiffening (isolated systolic hypertension in older adults), obstructive sleep apnea (↑ sympathetic tone), high-output state (eg, anemia, hyperthyroidism), exercise (transient). ↓ PP in aortic stenosis, cardiogenic shock, cardiac tamponade, advanced HF.
Mean arterial pressure	MAP = CO × total peripheral resistance (TPR)	MAP (at resting HR) = 2/3 DBP + 1/3 SBP = DBP + 1/3 PP.

Starling curves



Force of contraction is proportional to end-diastolic length of cardiac muscle fiber (preload).

↑ contractility with catecholamines, positive inotropes (eg, dobutamine, milrinone, digoxin).

↓ contractility with loss of functional myocardium (eg, MI), β-blockers (acutely), nondihydropyridine Ca^{2+} channel blockers, HF.

Resistance, pressure, flow

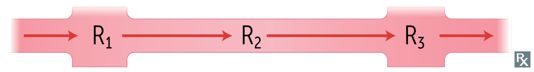
Volumetric flow rate ($\dot{Q}$) = flow velocity (v) × cross-sectional area (A)

Resistance (R)

$$= \frac{\text{driving pressure } (\Delta P)}{\dot{Q}} = \frac{8\eta \text{ (viscosity)} \times \text{length}}{\pi r^4}$$

Total resistance of vessels in series:

$R_T = R_1 + R_2 + R_3 \ldots$



Total resistance of vessels in parallel:

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \ldots$$

$\Delta P = \dot{Q} \times R$ is analogous to Ohm's Law for electrical circuits ($V = I \times R$).

$\dot{Q} \propto r^4$

$R \propto 1/r^4$

Capillaries have highest total cross-sectional area and lowest flow velocity.

Pressure gradient drives flow from high pressure to low pressure.

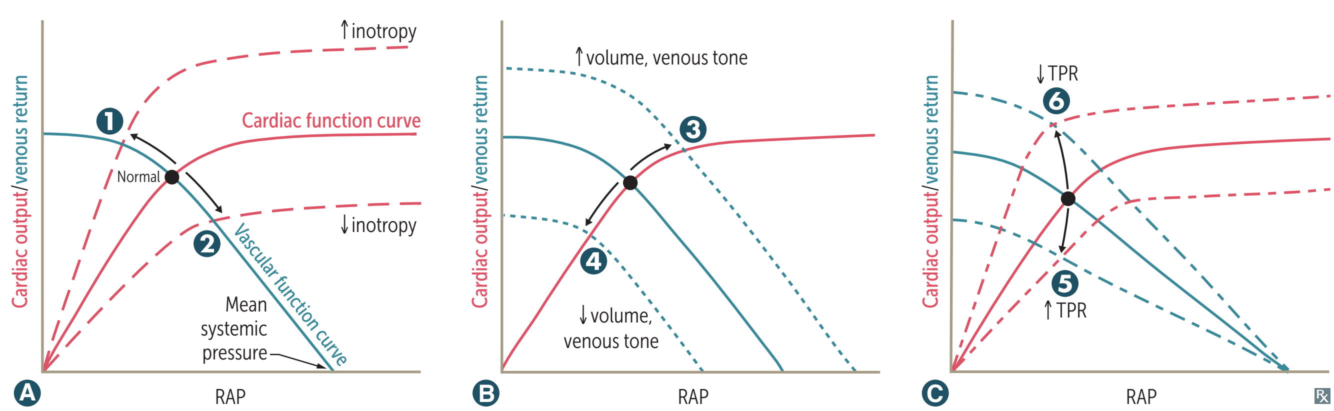
Arterioles account for most of TPR. Veins provide most of blood storage capacity.

Viscosity depends mostly on hematocrit.

Viscosity ↑ in hyperproteinemic states (eg, multiple myeloma), polycythemia.

Viscosity ↓ in anemia.

Cardiac and vascular function curves

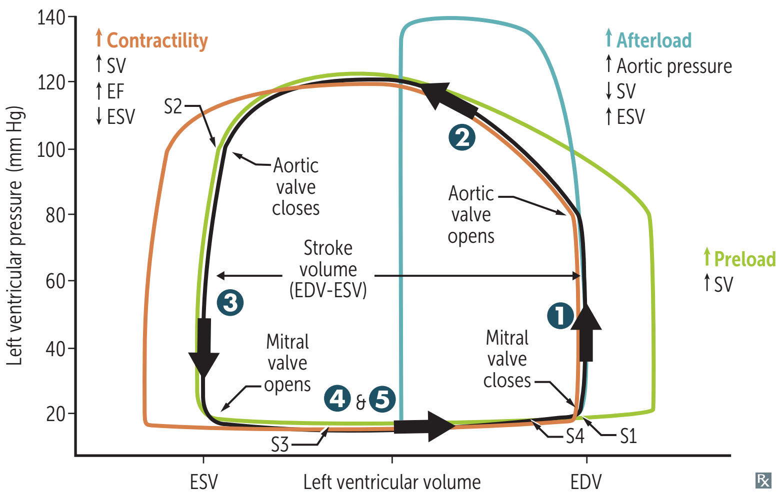


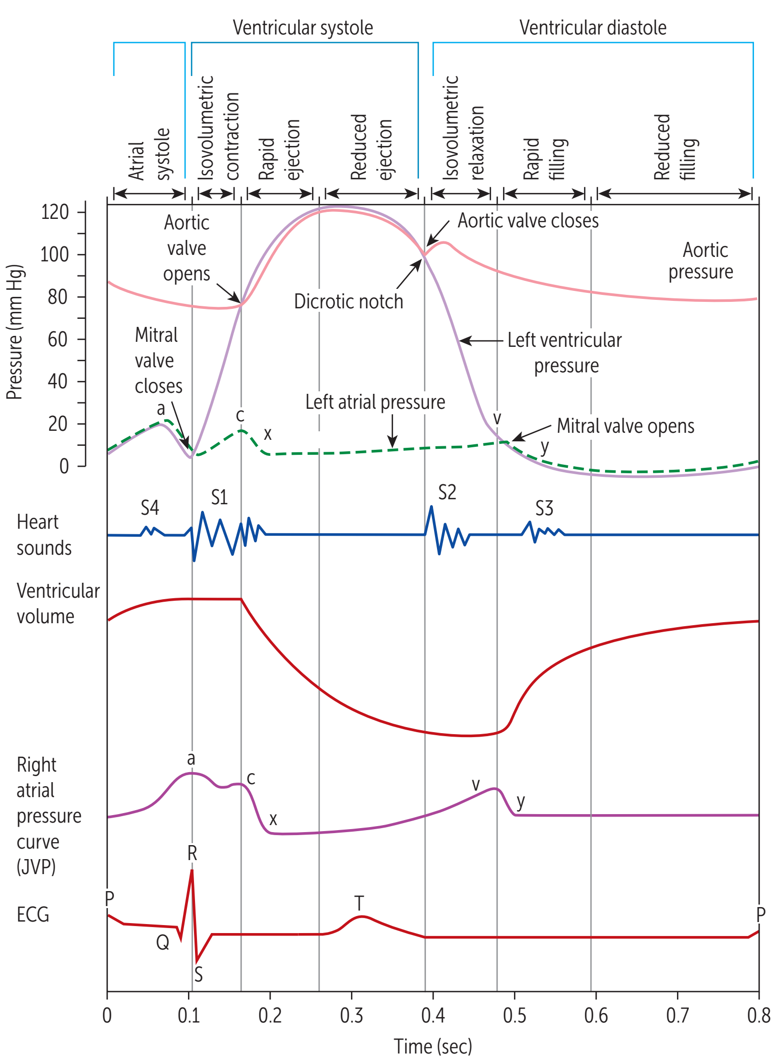
Intersection of curves = operating point of heart (ie, venous return and CO are equal, as circulatory system is a closed system).

GRAPH	EFFECT	EXAMPLES
Ⓐ **Inotropy**	Changes in contractility → altered SV → altered CO/venous return (VR) and RA pressure (RAP)	❶ Catecholamines, dobutamine, digoxin, exercise ⊕ ❷ HF with reduced EF, narcotic overdose, sympathetic inhibition ⊖
Ⓑ **Venous return**	Changes in circulating volume → altered RAP → altered SV → change in CO	❸ Fluid infusion, sympathetic activity, arteriovenous shunt ⊕ ❹ Acute hemorrhage, spinal anesthesia ⊖
Ⓒ **Total peripheral resistance**	Changes in TPR → altered CO Change in RAP unpredictable	❺ Vasopressors ⊕ ❻ Exercise, arteriovenous shunt ⊖

Changes often occur in tandem, and may be reinforcing (eg, exercise ↑ inotropy and ↓ TPR to maximize CO) or compensatory (eg, HF ↓ inotropy → fluid retention to ↑ preload to maintain CO).

Pressure-volume loops and cardiac cycle





The black loop represents normal cardiac physiology.

Phases—left ventricle:

1. Isovolumetric contraction—period between mitral valve closing and aortic valve opening; period of highest O_2 consumption
2. Systolic ejection—period between aortic valve opening and closing
3. Isovolumetric relaxation—period between aortic valve closing and mitral valve opening
4. Rapid filling—period just after mitral valve opening
5. Reduced filling—period just before mitral valve closing

Heart sounds:

S1—mitral and tricuspid valve closure. Loudest at mitral area.

S2—aortic and pulmonary valve closure. Loudest at left upper sternal border.

S3—in early diastole during rapid ventricular filling phase. Best heard at apex with patient in left lateral decubitus position. Associated with ↑ filling pressures (eg, MR, AR, HF, thyrotoxicosis) and more common in dilated ventricles (but can be normal in children, young adults, athletes, and pregnancy). Turbulence caused by blood from LA mixing with ↑ ESV.

S4—in late diastole (“atrial kick”). Turbulence caused by blood entering stiffened LV. Best heard at apex with patient in left lateral decubitus position. High atrial pressure. Associated with ventricular noncompliance (eg, hypertrophy). Considered abnormal if palpable. Common in older adults.

Jugular venous pulse (JVP):

a wave—**a**trial contraction. Prominent in AV dissociation (cannon a wave) and ↑ RV end-diastolic pressure from any cause. Absent in atrial fibrillation.

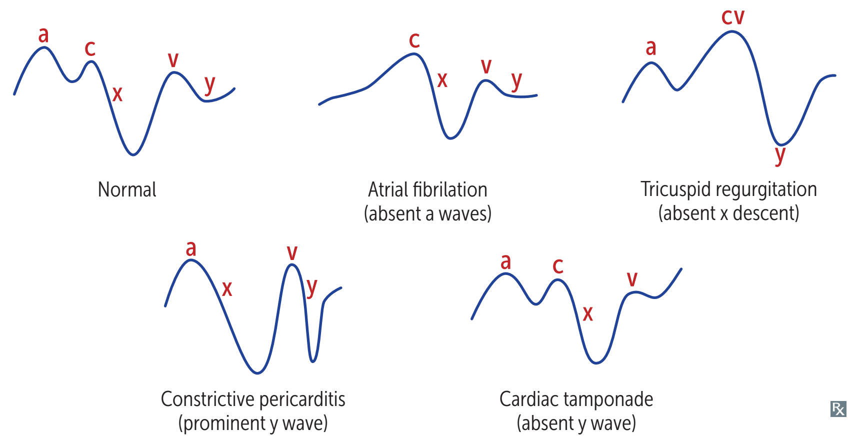
c wave—RV **c**ontraction (**c**losed tricuspid valve bulging into atrium).

x descent—atrial rela**x**ation and downward displacement of closed tricuspid valve during rapid ventricular ejection phase. Reduced or absent in tricuspid regurgitation and right HF because pressure gradients are reduced.

v wave—↑ RA pressure due to ↑ **v**olume against closed tricuspid **v**alve.

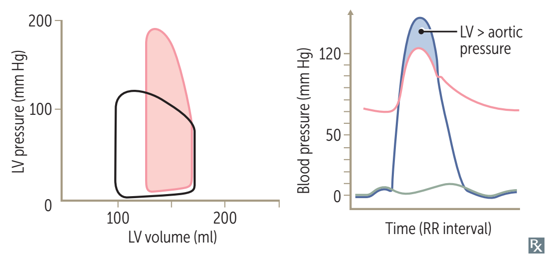
y descent—RA empt**y**ing into RV. Prominent in constrictive pericarditis, absent in cardiac tamponade.

Jugular venous pressure tracings



Pressure-volume loops and valvular disease

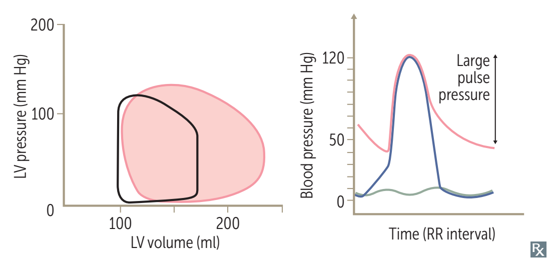
Aortic stenosis



↑ LV pressure
↑ ESV
No change in EDV (if mild)
↓ SV

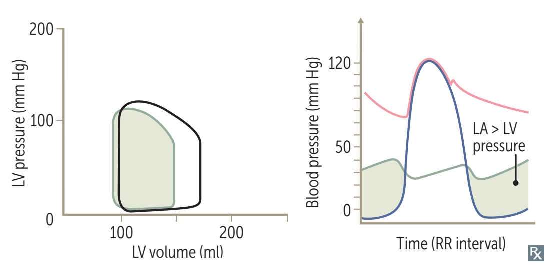
Ventricular hypertrophy → ↓ ventricular compliance → ↑ EDP for given EDV

Aortic regurgitation



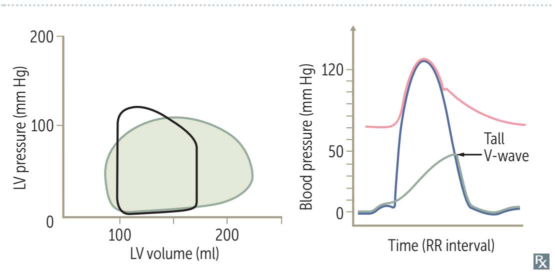
No true isovolumetric phase
↑ EDV
↑ SV
Loss of dicrotic notch

Mitral stenosis



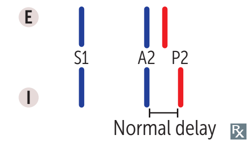
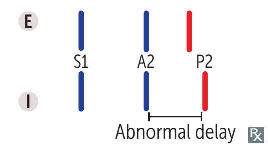
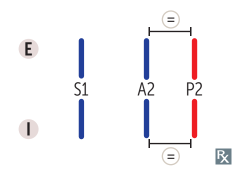
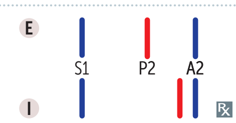
↑ LA pressure
↓ EDV because of impaired ventricular filling
↓ ESV
↓ SV

Mitral regurgitation

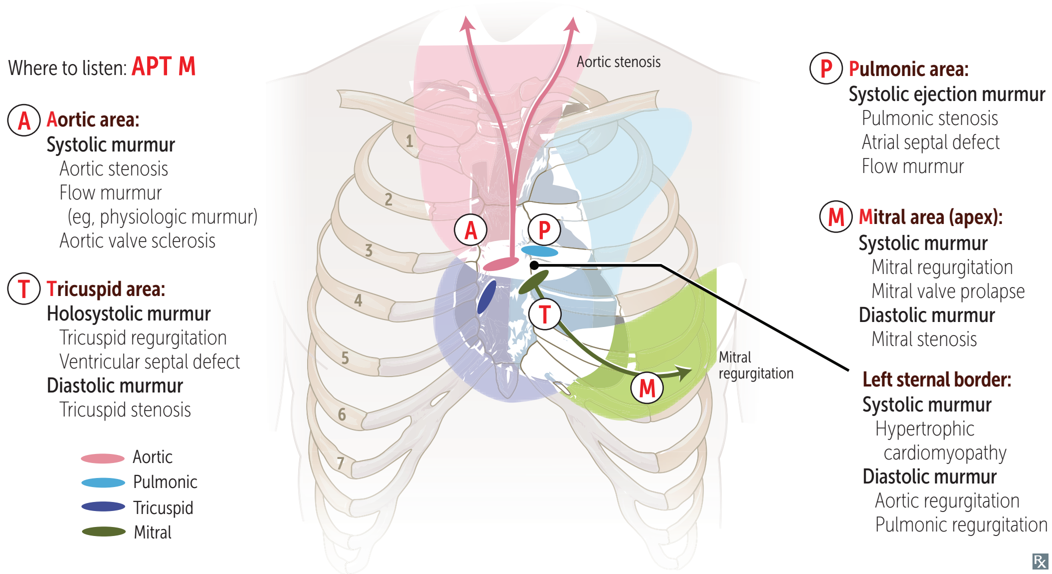


No true isovolumetric phase
↓ ESV due to ↓ resistance and ↑ regurgitation into LA during systole
↑ EDV due to ↑ LA volume/pressure from regurgitation → ↑ ventricular filling
↑ SV (forward flow into systemic circulation plus backflow into LA)

Splitting of S2

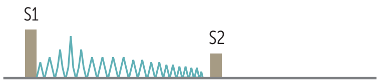
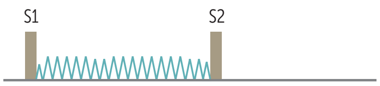
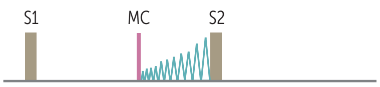
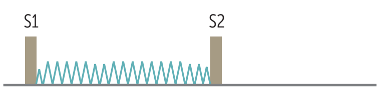
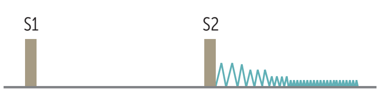
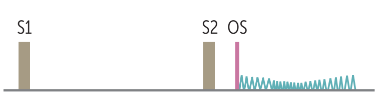
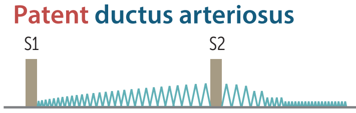
Physiologic splitting	Inspiration → drop in intrathoracic pressure → ↑ venous return → ↑ RV filling → ↑ RV stroke volume → ↑ RV ejection time → delayed closure of pulmonic valve. ↓ pulmonary impedance (↑ capacity of the pulmonary circulation) also occurs during inspiration, which contributes to delayed closure of pulmonic valve.	 E = Expiration I = Inspiration
Wide splitting	Seen in conditions that delay RV emptying (eg, pulmonic stenosis, right bundle branch block). Causes delayed pulmonic sound (especially on inspiration). An exaggeration of normal splitting.	
Fixed splitting	Heard in ASD. ASD → left-to-right shunt → ↑ RA and RV volumes → ↑ flow through pulmonic valve → delayed pulmonic valve closure (independent of respiration).	
Paradoxical splitting	Heard in conditions that delay aortic valve closure (eg, aortic stenosis, left bundle branch block). Normal order of semilunar valve closure is reversed: in **pa**radoxical splitting **P**2 occurs before **A**2. On inspiration, P2 closes later and moves closer to A2, "paradoxically" eliminating the split. On expiration, the split can be heard (opposite to physiologic splitting).	

Auscultation of the heart



MANEUVER	CARDIOVASCULAR CHANGES	MURMURS THAT INCREASE WITH MANEUVER	MURMURS THAT DECREASE WITH MANEUVER
Standing, Valsalva (strain phase)	↓ preload (↓ LV volume)	MVP (↓ LV volume) with earlier midsystolic click HCM (↓ LV volume)	Most murmurs (↓ flow through stenotic or regurgitant valve)
Passive leg raise	↑ preload (↑ LV volume)	Most murmurs (↑ flow through stenotic or regurgitant valve)	MVP (↑ LV volume) with later midsystolic click HCM (↑ LV volume)
Squatting	↑ preload, ↑ afterload (↑ LV volume)		
Hand grip	↑↑ afterload → ↑ reverse flow across aortic valve (↑ LV volume)	Most other left-sided murmurs (AR, MR, VSD)	AS (↓ transaortic valve pressure gradient) HCM (↑ LV volume)
Inspiration	↑ venous return to right heart, ↓ venous return to left heart	Most right-sided murmurs (increase with inspiration)	Most left-sided murmurs (increase with expiration)

Heart murmurs

	AUSCULTATION	CLINICAL ASSOCIATIONS	NOTES
Systolic			
Aortic stenosis 	Crescendo-decrescendo ejection murmur, loudest at heart base, radiates to carotids Soft S2 +/– ejection click "Pulsus parvus et tardus"—weak pulses with delayed peak	Age-related calcification (> 60 years old) Early-onset calcification of bicuspid aortic valve (~ 50–60 years old)	Can lead to **S**yncope, **A**ngina, **D**yspnea on exertion (**SAD**) LV pressure > aortic pressure during systole
Mitral/tricuspid regurgitation 	Holosystolic, high-pitched "blowing" murmur MR: loudest at apex, radiates toward axilla TR: loudest at tricuspid area	MR: often due to ischemic heart disease (post-MI), MVP, LV dilatation, rheumatic fever (RF) TR: often due to RV dilatation; may be 2° to permanent pacemaker placement MR or TR: infective endocarditis	
Mitral valve prolapse 	Late crescendo murmur with midsystolic click (MC) that occurs after carotid pulse Best heard over apex Loudest just before S2	Usually benign, but can predispose to infective endocarditis Can be caused by RF, chordae rupture, mitral annular disjunction, or myxomatous degeneration (1° or 2° to connective tissue disease)	MC due to sudden tensing of chordae tendineae as mitral leaflets prolapse into LA (**c**hordae cause **c**rescendo with **c**lick)
Ventricular septal defect 	Holosystolic, harsh-sounding murmur Loudest at tricuspid area	Congenital	Larger VSDs have lower intensity murmur than smaller VSDs
Diastolic			
Aortic regurgitation 	Early diastolic, decrescendo, high-pitched "blowing" murmur best heard at base (aortic root dilation) or left sternal border (valvular disease)	Causes include **BEAR**: **B**icuspid aortic valve, **E**ndocarditis, **A**ortic root dilation, **R**F Wide pulse pressure, pistol shot femoral pulse, pulsing nail bed	Hyperdynamic pulse and head bobbing when severe and chronic Can progress to left HF
Mitral stenosis 	Follows opening snap (OS) Delayed rumbling mid-to-late murmur (↓ interval between S2 and OS correlates with ↑ severity)	Late and highly specific sequelae of RF Chronic MS can result in LA dilation and pulmonary congestion, atrial fibrillation, Ortner syndrome, hemoptysis, right HF	OS due to abrupt halt in leaflet motion in diastole after rapid opening due to fusion at leaflet tips LA >> LV pressure during diastole
Continuous			
Patent ductus arteriosus 	Continuous **machine**like murmur, best heard at left infraclavicular area Loudest at S2	Often due to congenital rubella or prematurity	You need a **patent** for that **machine**.

Myocardial action potential

Phase 0 = rapid upstroke and depolarization—fast voltage-gated Na^+ channels open.

Phase 1 = initial repolarization—inactivation of voltage-gated Na^+ channels. Transient outward voltage-gated K^+ channels begin to open.

Phase 2 = plateau ("pla**two**")—Ca^{2+} influx through voltage-gated Ca^{2+} channels balances K^+ efflux. Ca^{2+} influx triggers Ca^{2+} release from sarcoplasmic reticulum and myocyte contraction (excitation-contraction coupling).

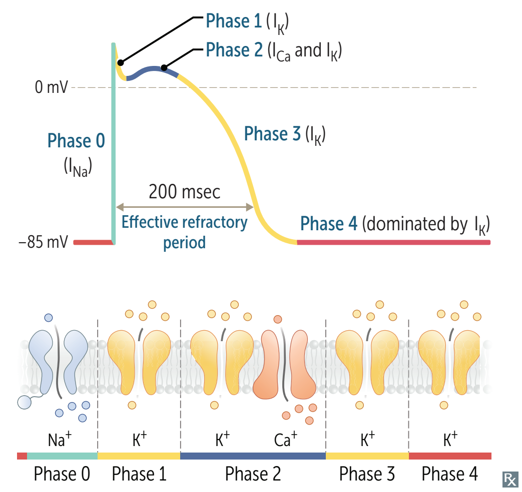
Phase 3 = rapid repolarization—K^+ efflux due to opening of voltage-gated slow delayed-rectifier K^+ channels and closure of voltage-gated Ca^{2+} channels.

Phase 4 = resting potential—high K^+ permeability through K^+ channels.

In contrast to skeletal muscle, cardiac muscle has the following characteristics:

- Action potential has a plateau due to Ca^{2+} influx and opposing K^+ efflux.
- Contraction requires Ca^{2+} influx from ECF to induce Ca^{2+} release from sarcoplasmic reticulum (Ca^{2+}-induced Ca^{2+} release).
- Myocytes conduct excitation throughout the heart via gap junctions.

Occurs in all cardiac myocytes except for those in the SA and AV nodes.



Pacemaker action potential

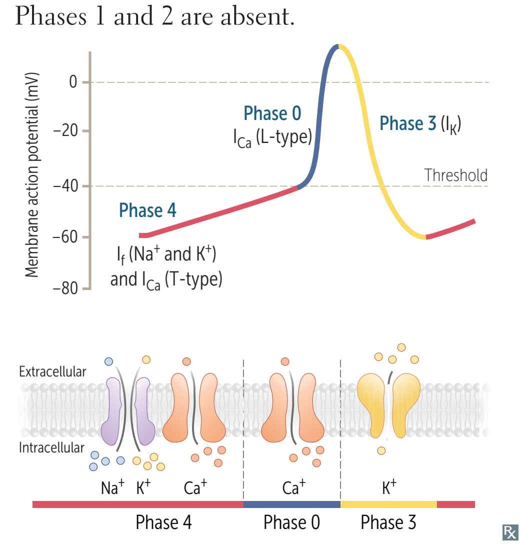
Key differences from the ventricular action potential include:

Phase 4 = slow spontaneous diastolic depolarization due to I_f ("funny current"). HCN channels are responsible for a slow, mixed Na^+/K^+ inward current; different from I_{Na} in phase 0 of ventricular action potential. Accounts for automaticity of SA and AV nodes. The slope of phase 4 in the SA node determines HR. ACh/adenosine ↓ the rate of diastolic depolarization and ↓ HR, while catecholamines ↑ depolarization and ↑ HR. Sympathetic stimulation ↑ the chance that HCN channels are open and thus ↑ HR.

Phase 0 = upstroke—opening of voltage-gated Ca^{2+} channels. Fast voltage-gated Na^+ channels are permanently inactivated due to the less negative resting potential of these cells → slow conduction velocity, used by AV node to prolong transmission from atria to ventricles.

Phase 3 = repolarization—inactivation of Ca^{2+} channels and ↑ activation of K^+ channels → ↑ K^+ efflux.

Occurs in the SA and AV nodes. Phases 1 and 2 are absent.



Electrocardiogram

Conduction pathway: SA node → atria → AV node → bundle of His → right bundle branch and left bundle branch (which divides into left anterior and posterior fascicles) → Purkinje fibers → ventricles; left bundle branch divides into left anterior and posterior fascicles.

SA node—located in upper crista terminalis near SVC; typically serves as dominant "pacemaker" for HR, with slow phase of upstroke.

AV node—located in interatrial septum near coronary sinus opening. Blood supply via PDA. 100-msec delay allows time for ventricular filling.

Pacemaker rates: SA > atria > AV > bundle of His/Purkinje/ventricles.

Speed of conduction: **H**is-**P**urkinje > **At**ria > **Vent**ricles > **AV** node. **H**e **P**arks **At** **Vent**ura **AV**enue.

P wave—atrial depolarization.

PR interval—time from start of atrial depolarization to start of ventricular depolarization (normally 120-200 msec).

QRS complex—ventricular depolarization (normally ≤ 100 msec).

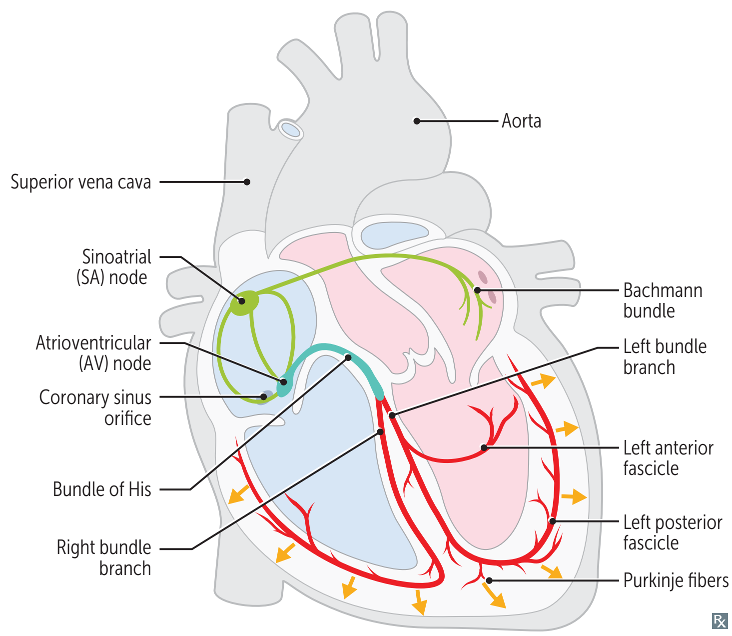
QT interval—ventricular depolarization, contraction, repolarization.

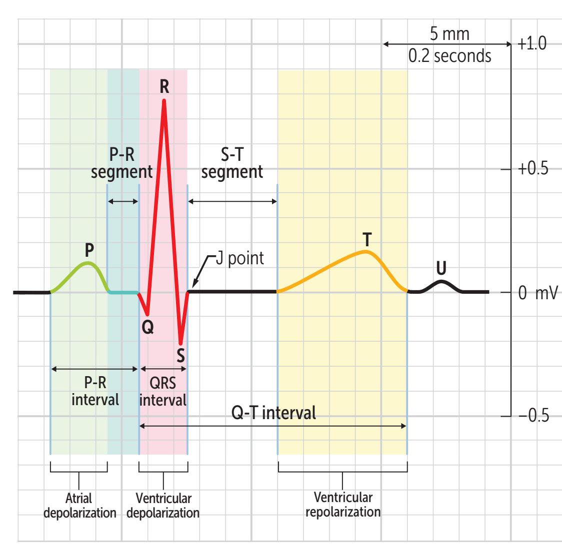
T wave—ventricular repolarization. Inversion may indicate ischemia or recent MI.

J point—junction between end of QRS complex and start of ST segment.

ST segment—isoelectric, ventricles depolarized.

U wave—prominent in hypokalemia (think hyp"**U**"kalemia), bradycardia.





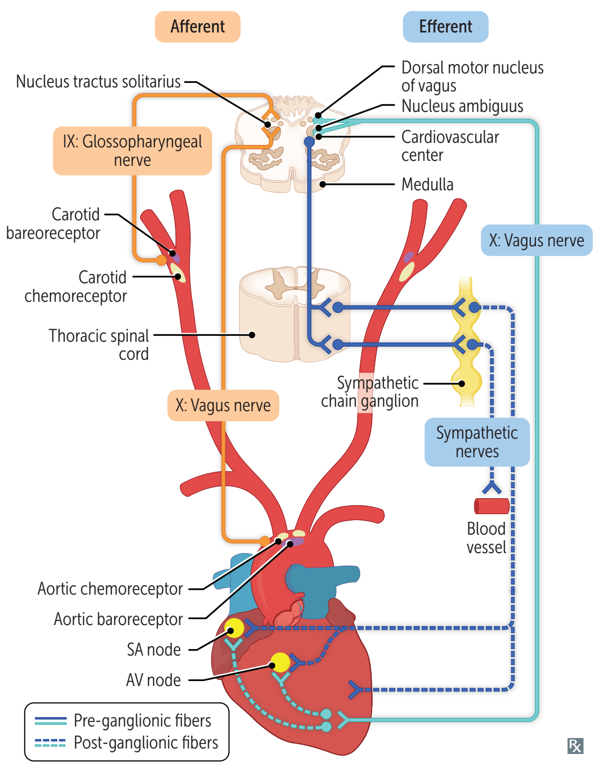
Atrial natriuretic peptide

Released from **atrial myocytes** in response to ↑ blood volume and atrial pressure. Acts via cGMP. Causes vasodilation and ↓ Na^+ reabsorption at the renal collecting tubule. Dilates afferent renal arterioles and constricts efferent arterioles, promoting diuresis and contributing to "aldosterone escape" mechanism.

B-type (brain) natriuretic peptide

Released from **ventricular myocytes** in response to ↑ tension. Similar physiologic action to ANP, with longer half-life. BNP blood test used for diagnosing HF (very good negative predictive value).

Baroreceptors and chemoreceptors



Receptors:

- Aortic arch transmits via vagus nerve to nucleus tractus solitarius of medulla (responds to changes in BP).
- Carotid sinus (dilated region superior to bifurcation of carotid arteries) transmits via glossopharyngeal nerve to nucleus tractus solitarius of medulla (responds to changes in BP).

Chemoreceptors:

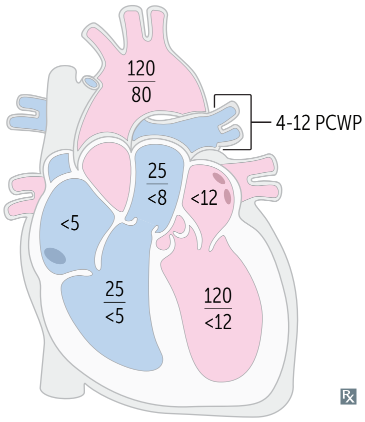
- Peripheral—carotid and aortic bodies are stimulated by ↑ P_{CO_2}, ↓ pH of blood, and ↓ P_{O_2} (< 60 mm Hg).
- Central—are stimulated by changes in pH and P_{CO_2} of brain interstitial fluid, which in turn are influenced by arterial CO_2 as H^+ cannot cross the blood-brain barrier. Do not respond to P_{O_2}. Central chemoreceptors become less sensitive with chronically ↑ P_{CO_2} (eg, COPD) → ↑ dependence on peripheral chemoreceptors to detect ↓ O_2 to drive respiration.

Baroreceptors:

- Hypotension—↓ arterial pressure → ↓ stretch → ↓ afferent baroreceptor firing → ↑ efferent sympathetic firing and ↓ efferent parasympathetic stimulation → vasoconstriction, ↑ HR, ↑ contractility, ↑ BP. Important in the response to hypovolemic shock.
- Carotid sinus hypersensitivity—elicited by carotid massage, shaving, tight necktie or shirt collar → ↑ carotid sinus pressure → ↑ afferent baroreceptor firing → ↑ AV node refractory period → ↓ HR → ↓ CO. Also leads to peripheral vasodilation. Can cause presyncope/syncope. Exaggerated in underlying atherosclerosis, prior neck surgery, older age.
- Component of Cushing reflex (triad of hypertension, bradycardia, and respiratory depression)—↑ intracranial pressure constricts arterioles → cerebral ischemia → ↑ pCO_2 and ↓ pH → central reflex sympathetic ↑ in perfusion pressure (hypertension) → ↑ stretch → peripheral reflex baroreceptor–induced bradycardia.

Normal resting cardiac pressures

Pulmonary capillary wedge pressure (PCWP; in mm Hg) is a good approximation of left atrial pressure, except in mitral stenosis when PCWP > LV end diastolic pressure. PCWP is measured with pulmonary artery catheter (Swan-Ganz catheter).



Autoregulation

How blood flow to an organ remains constant over a wide range of perfusion pressures.

ORGAN	FACTORS DETERMINING AUTOREGULATION	
Lungs	Hypoxia causes vasoconstriction	The pulmonary vasculature is unique in that alveolar hypoxia causes vasoconstriction so that only well-ventilated areas are perfused. In other organs, hypoxia causes vasodilation
Heart	Local metabolites (vasodilatory): NO, CO_2, ↓ O_2	
Brain	Local metabolites (vasodilatory): CO_2 (pH)	
Kidneys	Myogenic (stretch-dependent response of afferent arteriole) and tubuloglomerular feedback	
Skeletal muscle	Local metabolites during exercise (vasodilatory): CO_2, H^+, Adenosine, Lactate, K^+ At rest: sympathetic tone in arteries	CHALK
Skin	Sympathetic vasoconstriction most important mechanism for temperature control	

Capillary fluid exchange

Starling forces determine fluid movement through capillary walls:

- P_c = capillary hydrostatic pressure—pushes fluid out of capillary
- P_i = interstitial hydrostatic pressure—pushes fluid into capillary
- π_c = plasma oncotic pressure—pulls fluid into capillary
- π_i = interstitial fluid oncotic pressure—pulls fluid out of capillary

J_v = net fluid flow = $K_f [(P_c - P_i) - \sigma(\pi_c - \pi_i)]$

K_f = capillary permeability to fluid

σ = reflection coefficient (measure of capillary impermeability to protein)

Edema—excess fluid outflow into interstitium commonly caused by:

- ↑ capillary hydrostatic pressure (↑ P_c; eg, HF)
- ↑ capillary permeability (↑ K_f; eg, toxins, infections, burns)
- ↑ interstitial fluid oncotic pressure (↑ π_i; eg, lymphatic blockage)
- ↓ plasma proteins (↓ π_c; eg, nephrotic syndrome, liver failure, protein malnutrition)